

NIVEDH SUNIL

Backend AI Engineer · Full-Stack Developer · Systems Builder

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PROFESSIONAL SUMMARY

Third-year CSE student and Backend AI Engineering Intern at FlyRank AI. Built production agentic backends (FastAPI + LLMs), a bare-metal OS in C, and a full transformer from scratch in Zig. Seeking full-time Backend AI Engineer, Backend Engineer, and Full-Stack Developer roles.

EXPERIENCE

Backend AI Engineering Intern — FlyRank AI · Bengaluru

2026 – Present

- ▶ Shipped agentic AI backend systems to production in Python and FastAPI, integrating LLM reasoning into internal tooling used across the AI Engineering cohort
- ▶ Designed and built REST APIs and backend infrastructure end-to-end for cohort-based capstone deliverables
- ▶ Worked directly with LLM integration pipelines, translating cohort project specs into shipped backend services

FEATURED PROJECTS

N-OS — Bare-Metal Operating System — C · Assembly · x86

2025 – Present

- ▶ Built a 32-bit OS from scratch in C — bootloader, kernel, GUI, virtual file system, and networking stack — with no host OS underneath
- ▶ Wrote PE and ELF binary loaders from scratch, enabling native execution of both Windows and Linux binaries inside N-OS
- ▶ Hand-rolled a full TCP/IP stack (RTL8139 driver, IPv4, 3-way handshake) that fetches live HTTP payloads; ships as a bootable .iso running in VirtualBox on 64MB RAM

N.E.O.S — Autonomous AI Orchestrator — Python · PyQt6 · Groq API · ChromaDB

2026

- ▶ Built an autonomous AI orchestrator with hybrid SQLite + ChromaDB memory and real-time Voice Activity Detection for hands-free interaction
- ▶ Engineered JSON-based agentic tool-calling on Groq's Llama 3.3 70B, mapping natural language to shell, Python, and system-level actions

ZigNGPT v2 — Transformer from Scratch — Zig · Systems Programming · ML

Jun 2025 – Present

- ▶ Built a language model from zero in Zig, replacing a v1 trigram Markov model with a full transformer architecture
- ▶ Hand-wrote attention, positional encoding, and backpropagation at the matrix level — zero ML framework dependencies

MAT-CHA.AI — Startup × Creator Matchmaking — FastAPI · React · AWS · DynamoDB · Bedrock

Jan – Mar 2026

- ▶ Built an AI-driven matchmaking platform bridging creators and startups at the AWS 'Build For India' Hackathon
- ▶ Architected a serverless AWS pipeline (S3, Lambda, DynamoDB, Bedrock embeddings, Amplify) and shipped it live within the hackathon's time constraints
- ▶ Extended the platform with an MCP tool server exposing matchmaking functions to any MCP-compatible LLM client

ADDITIONAL PROJECTS

- ▶ **Civic Twin — Bangalore Digital Assistant (Node.js, AWS Bedrock, Elasticsearch, Supabase)** — Bedrock-orchestrated Llama 3.3 assistant unifying 3 civic services (BBMP, BESCOM, BWSSB), with an Elasticsearch knowledge base and family-delegation access — built at the Elastic × AWS UG Bengaluru Hacknight.
- ▶ **Buying-Signal Scanner (Node.js, Browser-Use)** — GTM browser-automation agent that scans live web sources for buying-intent signals to feed sales pipelines — Browser-Use Hackathon.
- ▶ **LabMate — Medical AI Report Simplifier (Next.js, LLaMA 3, FastAPI)** — Translates medical PDFs into plain-English explanations with lifestyle suggestions; deployed live as an installable PWA.
- ▶ **World Cup 2026 Operations Center (React, FastAPI)** — Live operations dashboard for real-time World Cup 2026 event tracking, deployed and publicly accessible.
- ▶ **TERRA-X — Real-Time Global Scenario Lab (FastAPI, React Three Fiber, Groq Llama 3.3)** — Interactive 360° 3D globe simulating AI-projected future outcomes for any city from user-input coordinates.
- ▶ **Emotion-Aware Debugging Assistant (Python, OpenCV, MediaPipe, DeepFace)** — Real-time CV system tracking blink rate, posture, and facial emotion to compute a live per-file developer frustration score.
- ▶ **Terr-AR-ium — 3D Carbon Footprint Tracker (React Three Fiber, Groq Llama 3.1)** — Gamified carbon tracker where a voxel island grows or decays with real eco habits — built at Google Hack2Skill.
- ▶ **NGPT — Hybrid LLM Chatbot (LLaMA 2, GPT-Neo, FastAPI, LangChain)** — Routes queries between LLaMA 2 and a GPT-Neo retrieval-QA pipeline, with web search and voice input/output.

- ▶ **Smart Mobility Companion (Python, GPT-2, LLaMA, Hugging Face)** — Generative AI system for Bangalore bus routing, fare estimation, and timeline insights.

TECHNICAL SKILLS

Languages: Python, JavaScript/Node.js, C, C++, Zig, Go, SQL

AI / ML: LLaMA 2/3, GPT-Neo, Groq, OpenCV, MediaPipe, DeepFace, PyTorch, Hugging Face, RAG, Agentic AI, MCP, Prompt Engineering

Web & Backend: FastAPI, React, React Three Fiber, Flask, Django REST, Node.js, REST APIs

Databases: PostgreSQL, MySQL, DynamoDB

Cloud & DevOps: AWS (Lambda, S3, DynamoDB, Bedrock, Amplify), Git, GitHub, CI/CD

EDUCATION

B.Tech, Computer Science & Engineering — Garden City University

Sep 2024 – Jun 2028

GPA: 9.55 · 9.55 · 9.82 · 9.125 across semesters | Coursework: Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Numerical Linear Algebra

Higher Secondary (Class XII) — Sri Chaitanya College of Education

Jun 2022 – May 2024

Secondary Education (Class X) — Jaigopal Garodia Rashtrorathana Vidya Kendra

May 2010 – May 2022

HACKATHONS & COMMUNITY

- ▶ Elastic × AWS UG Bengaluru Hacknight — built Civic Twin, an agentic digital assistant for Bangalore civic services
- ▶ Browser-Use Hackathon (GTM Edition) — built Buying-Signal Scanner, a browser automation agent for sales signal detection
- ▶ AWS 'Build For India' Hackathon — built MAT-CHA.AI, a live AI matchmaking platform for startups and creators
- ▶ Google Hack2Skill — built Terr-AR-ium, a 3D voxel carbon footprint gamification app
- ▶ MS Ramaiah Ideathon — built a Ransomware Readiness Assessment Tool as an adaptive SaaS (Top performer)
- ▶ SIH Internal Hackathon — competed and shipped a working prototype within constraints
- ▶ AWS Summit Bangalore (Technical Edition) — sessions on Lambda, S3, DynamoDB, Bedrock, and Amplify architecture
- ▶ ClickHouse Agentic AI Meetup — workshop with Co-Founder/CTO Alexey Milovidov
- ▶ AWS 'AI for Bharath' Hackathon — credentialed hackathon participation (AWS)
- ▶ Salesforce Agentforce 'AI Builders Day' Workshop — hands-on session on building AI agents with Agentforce
- ▶ IEEE Member, Garden City University Chapter — Dec 2024 – Jan 2026

CERTIFICATIONS

C++ Essentials 1 (Cisco) · JavaScript Essentials 1 (Cisco) · Automata & Computability (BITS Pilani/Coursera) · Learn ClickHouse OLAP DBMS (Udemy) · Database Management & Microsoft Fabric SQL (Udemy) · Design Thinking for Innovators (Coursera) · Introduction to Data Science (Infosys Springboard; Cisco) · Front-End Web Developer (Infosys Springboard) · AI For Everyone (DeepLearning.AI/Coursera) · Generative AI Mastery Workshop (NxtWave) · Graph Data Structures & Knowledge Graphs (Skillssoft) · Claude 101 (Anthropic) · Introduction to Claude Cowork (Anthropic) · AI Fluency Framework — Foundations, Builders, Educators, Nonprofits, Small Businesses, Students & Teaching (Anthropic)